We create chemistry

Safety Data Sheet
2EH Lights PA

1. Identification

Product identifier used on the label

2EH Lights PA

Recommended use of the chemical and restriction on use
Recommended use*: for industrial use only

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number

CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification
Chemical family: alcohol

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Flam. Liq.	3	Flammable liquids
Acute Tox.	4 (Inhalation - mist)	Acute toxicity
Skin Corr./Irrit.	2	Skin corrosion/irritation
Eye Dam./Irrit.	1	Serious eye damage/eye irritation
Skin Sens.	1	Skin sensitization
Repr.	2 (unborn child)	Reproductive toxicity
STOT SE	3 (Vapours may cause drowsiness and dizziness.)	Specific target organ toxicity — single exposure

STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity — single exposure
Aquatic Acute	3	Hazardous to the aquatic environment - acute

Label elements

Pictogram:

Signal Word:
Danger

Hazard Statement:

H226	Flammable liquid and vapour.
H318	Causes serious eye damage.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H336	May cause drowsiness or dizziness.
H335	May cause respiratory irritation.
H361	Suspected of damaging the unborn child.
H402	Harmful to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe mist or vapour.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P243	Take action to prevent static discharges.
P273	Avoid release to the environment.
P201	Obtain special instructions before use.
P241	Use explosion-proof electrical, ventilating and lighting equipment.
P202	Do not handle until all safety precautions have been read and understood.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.
P242	Use only non-sparking tools.
P240	Ground and bond container and receiving equipment.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P303 + P361 + P353	IF ON SKIN (or hair): Remove or Take off immediately all contaminated clothing. Rinse skin with water or shower.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P370 + P378	In case of fire: Use alcohol-resistant foam, carbon dioxide, dry powder or water spray for extinction.

Precautionary Statements (Storage):

P403 + P235	Store in a well-ventilated place. Keep cool.
P233	Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Hazards not otherwise classified

Labeling of special preparations (GHS):
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 0 - 1 %

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

n-butanol
CAS Number: 71-36-3
Content (W/W): 25.0 - <= 100.0%
Synonym: 1-Butanol; n-Butanol

Isobutanol
CAS Number: 78-83-1
Content (W/W): 3.0 - < 10.0%
Synonym: Isobutyl alcohol

2-ethylhexan-1-ol
CAS Number: 104-76-7
Content (W/W): 0.1 - < 50.0%
Synonym: 2-Ethylhexanol

2-ethylhexanal
CAS Number: 123-05-7
Content (W/W): 0.0 - < 3.0%
Synonym: No data available.

4-heptanone
CAS Number: 123-19-3
Content (W/W): 0.0 - < 3.0%
Synonym: Heptan-4-one

4. First-Aid Measures

Description of first aid measures

General advice:
Remove contaminated clothing.

If inhaled:
Remove the affected individual into fresh air and keep the person calm. Assist in breathing if necessary. If breathing difficulties develop, aid in breathing and seek immediate medical attention.

If on skin:
Remove contaminated clothing. Wash affected areas thoroughly with soap and water. Seek medical attention.

If in eyes:
Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes. Remove contact lenses, if present, after first 5 minutes, then continue rinsing. Seek medical attention.

If swallowed:
Rinse mouth and then drink 200-300 ml of water. Never induce vomiting or give anything by mouth if the victim is unconscious or having convulsions. Seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Overexposure may cause:, headache, dizziness, coordination disorder, coma, lacrimation, loss of hearing

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment:	Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, dry powder, alcohol-resistant foam, carbon dioxide

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
Vapours may form explosive mixture with air.

Advice for fire-fighters

Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Keep containers cool by spraying with water if exposed to fire. Foam should be applied in large quantities as it is broken down to some extent by the product.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Avoid inhalation. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Substance/product is RCRA hazardous due to its properties.

Discharge into the environment must be avoided.

Methods and material for containment and cleaning up
Spills should be contained, solidified, and placed in suitable containers for disposal.

7. Handling and Storage

Precautions for safe handling
Ensure thorough ventilation of stores and work areas.

Protection against fire and explosion:
Containers should be grounded against electrostatic charge. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities
No applicable information available.

Suitable materials for containers: Stove-lacquer KNS L-5X, Stove-lacquer Valspar HXR008F red, Carbon steel (Iron), tinned carbon steel (Tinplate), glass

Further information on storage conditions: Keep away from sources of ignition - No smoking. Keep container tightly closed in a cool, well-ventilated place. Storage containers should be grounded.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

n-butanol	OSHA PEL	PEL 100 ppm 300 mg/m3 ; SKIN_FINAL ; The substance can be absorbed through the skin. CLV 50 ppm 150 mg/m3 ;
	ACGIH TLV	TWA value 20 ppm ;
Isobutanol	OSHA PEL	PEL 100 ppm 300 mg/m3 ; TWA value 50 ppm 150 mg/m3 ;
	ACGIH TLV	TWA value 50 ppm ;
4-heptanone	OSHA PEL	TWA value 50 ppm 235 mg/m3 ;
	ACGIH TLV	TWA value 50 ppm ;

Advice on system design:
Provide local exhaust ventilation to maintain recommended P.E.L.

Personal protective equipment

Respiratory protection:
Observe OSHA regulations for respirator use (29 CFR 1910.134). Wear a NIOSH-certified (or equivalent) respirator as necessary.

Hand protection:
Chemical resistant protective gloves, Suitable materials, chloroprene rubber (Neoprene)

Eye protection:
Tightly fitting safety goggles (chemical goggles). Wear face shield if splashing hazard exists.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:
Work place should be equipped with a shower and an eye wash. Avoid inhalation of vapours/mists. Wear protective clothing as necessary to prevent contact.

9. Physical and Chemical Properties

Form:	liquid
Odour:	alcohol-like
Odour threshold:	No applicable information available.
Colour:	almost colourless
pH value:	7
Melting point:	No applicable information available.
Freezing point:	No data available.
Boiling range:	No applicable information available.
Boiling point:	119 °C (1,013 hPa) Information applies to the solvent.
Sublimation point:	No applicable information available.
Flash point:	31 - 35 °C
Flammability:	Flammable liquid
Vapour pressure:	7.33 hPa
Density:	0.8278 g/cm3
Relative density:	0.831
Vapour density:	No applicable information available.
Partitioning coefficient n-octanol/water (log Pow):	No applicable information available.
Thermal decomposition:	No applicable information available.
Viscosity, dynamic:	No applicable information available.
Viscosity, kinematic:	No applicable information available.
Solubility in water:	(25 °C) partly soluble
Solubility (quantitative):	No applicable information available.
Solubility (qualitative):	No applicable information available.
Evaporation rate:	No applicable information available.

10. Stability and Reactivity

Reactivity
No applicable information available.

Oxidizing properties:
not fire-propagating (other)

Chemical stability
No applicable information available.

Possibility of hazardous reactions
Reacts with strong oxidizing agents. Reacts with reducing agents. Reacts with sulfuric acid. Reacts with nitrites.

Conditions to avoid
Avoid all sources of ignition: heat, sparks, open flame.

Incompatible materials
No applicable information available.

Hazardous decomposition products

Decomposition products:
No applicable information available.

Thermal decomposition:
No applicable information available.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Harmful by inhalation. Virtually nontoxic after a single skin contact. Virtually nontoxic after a single ingestion.

Information on: n-butanol
Assessment of acute toxicity:Of low toxicity after short-term skin contact. Virtually nontoxic by inhalation. Of low toxicity after single ingestion. The European Union (EU) has classified this substance as 'harmful' after oral exposure.
If used as intended, this product is not expected to present a physical or health hazard.

Information on: 2-ethylhexan-1-ol
Assessment of acute toxicity:Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.
If used as intended, this product is not expected to present a physical or health hazard.
Of low toxicity after single ingestion. Of low toxicity after short-term skin contact. Of moderate toxicity after short-term inhalation.

Information on: 2-ethylhexanal
Assessment of acute toxicity:Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Information on: 4-heptanone
Assessment of acute toxicity:Of moderate toxicity after short-term inhalation. Of low toxicity after single ingestion. Of low toxicity after short-term skin contact.

Oral
Type of value: ATE
Value: > 5,000 mg/kg

Inhalation
Type of value: ATE
Value: > 20 mg/l
Determined for vapor

Type of value: ATE
Value: 5 mg/l
Determined for mist

Dermal
Type of value: ATE
Value: > 5,000 mg/kg

Assessment other acute effects
Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness). Causes temporary irritation of the respiratory tract.

Irritation / corrosion
Assessment of irritating effects: Causes serious eye damage. Causes skin irritation.

Information on: n-butanol
Assessment of irritating effects: Skin contact causes irritation. Risk of serious damage to eyes.

Information on: Isobutanol
Assessment of irritating effects: May cause severe damage to the eyes. Skin contact causes irritation.

Information on: 2-ethylhexan-1-ol
Assessment of irritating effects: Eye contact causes irritation. Skin contact causes irritation.

Information on: 4-heptanone
Assessment of irritating effects: Eye contact causes irritation. Skin contact causes irritation.

Sensitization
Assessment of sensitization: May cause allergic skin reaction.

Information on: 2-ethylhexanal
Assessment of sensitization:
sensitizing effect in animal tests

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: Based on available Data, the classification criteria are not met.

Genetic toxicity
Assessment of mutagenicity: Based on available Data, the classification criteria are not met.

Carcinogenicity
Assessment of carcinogenicity: Based on available Data, the classification criteria are not met.

Reproductive toxicity
Assessment of reproduction toxicity: Based on available Data, the classification criteria are not met.

Teratogenicity
Assessment of teratogenicity: May cause harm to the unborn child.

Information on: n-butanol
Assessment of teratogenicity: Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.
The potential to cause toxicity to development cannot be excluded when given in high doses.

Information on: 2-ethylhexanal
Assessment of teratogenicity: Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Other Information
The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
Harmful to aquatic life.

Aquatic toxicity

Information on: 2-ethylhexan-1-ol
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Additional information

Other ecotoxicological advice:
Acutely toxic for aquatic organisms. Do not discharge substance/product into sewer system. The product has not been tested. The statement has been derived from the properties of the individual components.

13. Disposal considerations

Waste disposal of substance:
Dispose of in a RCRA-licensed facility.
Do not discharge into waterways or sewer systems without proper authorization.

Container disposal:
Empty containers with less than 1 inch of residue may be landfilled at a licensed facility. Recommend crushing, puncturing or other means to prevent unauthorized use of used containers. If containers are not empty, they must be disposed of in a RCRA-licensed facility.

14. Transport Information

Land transport
USDOT

Hazard class:	3
Packing group:	III
ID number:	UN 1993
Hazard label:	3
Proper shipping name:	FLAMMABLE LIQUID, N.O.S. (contains N-BUTANOL, 2-METHYLPROPAN-1-OL)

Sea transport
IMDG

Hazard class:	3
Packing group:	III
ID number:	UN 1993
Hazard label:	3
Marine pollutant:	NO
Proper shipping name:	FLAMMABLE LIQUID, N.O.S. (contains N-BUTANOL, 2-METHYLPROPAN-1-OL)

Air transport
IATA/ICAO

Hazard class:	3
Packing group:	III
ID number:	UN 1993
Hazard label:	3
Proper shipping name:	FLAMMABLE LIQUID, N.O.S. (contains N-BUTANOL, 2-METHYLPROPAN-1-OL)

15. Regulatory Information

Federal Regulations

Registration status:
Chemical TSCA, US released / listed

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

EPCRA 313:

CAS Number	Chemical name
71-36-3	n-butanol

CERCLA RQ	CAS Number	Chemical name
5000 LBS	71-36-3; 78-83-1	n-butanol; Isobutanol
100 LBS	123-72-8	butyraldehyde
10 LBS	71-43-2	Benzene

State regulations

State RTK	CAS Number	Chemical name
NJ	71-36-3	n-butanol
	78-83-1	Isobutanol
	123-05-7	2-ethylhexanal
	123-19-3	4-heptanone
PA	71-36-3	n-butanol
	78-83-1	Isobutanol
	104-76-7	2-ethylhexan-1-ol
	123-05-7	2-ethylhexanal
	123-19-3	4-heptanone
	71-43-2	Benzene

Safe Drinking Water & Toxic Enforcement Act, CA Prop. 65:

WARNING: This product can expose you to chemicals including BENZENE, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

NFPA Hazard codes:
Health: 3 Fire: 3 Reactivity: 0 Special:

HMIS III rating
Health: 3 Flammability: 3 Physical hazard: 0

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2020/04/13

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